Lab: Flight Modes, Geofence, Return-to-Launch and Log Analysis in the ArduPilot Simulator

Course: U21MI504 – Unmanned Aerial Vehicle Technology · **Unit V** (topics 26, 27, 28 and 29) · **CO5**
Duration: 2 hours (one practical slot) · **Group size:** 2 (pilot and safety officer, swap halfway)
Tools: Mission Planner (Windows) with its built-in multirotor simulator, UAV Log Viewer (web). No aircraft and no hardware needed.

1. Why this lab

On a real drone, a failsafe test that goes wrong costs a crash. In software-in-the-loop (SITL) simulation the same ArduPilot firmware that runs on a flight controller runs on your laptop against a physics model. You can switch modes, break the GPS, cut the radio and drain the battery, then read the log to prove what the autopilot did and when.

By the end of the lab you will be able to:

1. Arm a simulated multirotor, take off and switch between Stabilize, AltHold, Loiter, Guided, Auto, RTL and Land, and explain what each mode holds for you.
2. Configure a geofence and the return-to-launch (RTL) altitude from the parameter list, and show the fence being enforced.
3. Trigger radio-loss, low-battery and GPS-loss events and record the autopilot's response.
4. Download the DataFlash log and use plots of the MODE, CTUN, BAT, GPS, ATT and VIBE messages to reconstruct the flight.

2. Before the lab (15 minutes at home)

- Install Mission Planner from the ArduPilot website (ardupilot.org, "Mission Planner" under Ground Control Stations). Windows only; Mac and Linux users pair with a Windows user or run QGroundControl with a separate SITL install (ask the instructor).
- Read the quick-start sheet *Mission Planner and QGroundControl Quick-Start* (same folder) and Chapter 5, sections on flight modes and failsafes.
- Bookmark UAV Log Viewer: **plot.ardupilot.org**.

Version note. ArduPilot renames some simulator parameters between releases. Where this sheet gives a parameter name, search the parameter list for the first part (for example `SIM_GPS`) if the exact name is missing.

3. Safety-officer rules, even in simulation

The safety officer reads each step aloud, confirms the pilot has done it and writes the time on the worksheet. Treat the simulator like the field: announce "arming", "taking off", "mode change to …" and "landing". This is the habit you will need on the real aircraft.

4. Part A: Start the simulator and arm (15 min)

1. Open Mission Planner and click **SIMULATION** in the top menu.
2. Choose **Multirotor**. Accept the stable firmware if asked. Mission Planner downloads the simulator and connects automatically. The default home location is an airfield in Australia; that is normal.
3. Wait until the HUD shows a GPS fix (3D fix or better) and the **Messages** tab stops listing pre-arm errors. Write the number of satellites and the HDOP from the **Status** tab on the worksheet.
4. In **Flight Data → Actions**, select mode **GUIDED** and click **Set Mode**.
5. Click **Arm/Disarm**. The HUD should change from DISARMED to ARMED.
6. Right-click on the map near the vehicle, choose **Takeoff**, and enter **20 m**. The vehicle climbs to 20 m and holds.

Checkpoint A. Altitude on the HUD within 1 m of 20 m. Note the battery voltage at hover.

5. Part B: Flight modes, predict then observe (25 min)

If no joystick or gamepad is connected, the simulated throttle stays at its default (normally low). This makes the differences between modes very visible. For each row, **write your prediction first**, then switch mode from the Actions tab, watch for 20 seconds, and record what happened.

Step	Switch to	What does the mode hold for you?	Prediction with throttle low	What you observed
B1	LOITER	Altitude and horizontal position (needs GPS)		
B2	GUIDED (then Takeoff to 20 m again if needed)	Position commanded by the ground station		
B3	ALT_HOLD	Altitude only; drifts with wind		
B4	GUIDED, re-climb to 20 m			
B5	STABILIZE	Attitude only; pilot owns throttle		

> **Expected behavior (read after predicting).** In Loiter and AltHold, low throttle is a *descend* command, so the vehicle comes down at a controlled rate. In Stabilize, low throttle means low motor output, so the vehicle drops much faster. If it lands or crashes, switch to GUIDED, arm and take off again. That is the point of simulation.

If your group has a USB gamepad, set it up in **Flight Data → Actions → Joystick**, center the throttle, and repeat B1, B3 and B5 while flying the sticks. Note how much more work Stabilize is.

Checkpoint B. Name the lowest mode on the ladder that still works with no GPS, and the first mode that stops working when GPS is lost.

6. Part C: Geofence and RTL altitude (20 min)

Open **CONFIG → Full Parameter List**, type each name in the search box, change the value, then click **Write Params**. Click **Refresh Params** and confirm the values read back.

Parameter	Set to	Meaning
`FENCE_ENABLE`	1	Fence on
`FENCE_TYPE`	3	Bitmask: 1 = maximum altitude, 2 = circle; 3 = both
`FENCE_ALT_MAX`	40	Ceiling, meters above home
`FENCE_RADIUS`	120	Circle radius, meters from home
`FENCE_ACTION`	1	On breach: RTL, or Land if RTL is not possible
`RTL_ALT`	3000	RTL climb altitude in **centimeters** (30 m)

1. In GUIDED at 20 m, right-click a point about 200 m from home and choose **Fly To Here**. Watch the fence trigger.
2. Record: distance from home at breach, mode after breach, altitude the vehicle climbed or descended to before heading home, and where it landed.
3. Take off again to 20 m and use **Fly To Here** towards a point inside the fence, then use the Actions tab to command a climb (Change Alt) to 50 m. Record what happens at 40 m.

Checkpoint C. Explain why `RTL_ALT` is in centimeters and `FENCE_ALT_MAX` in meters, and why mixing them up is dangerous on a real site with a 22 m tree.

7. Part D: Failure injection (25 min)

Do one failure at a time. Before each one, take off to 20 m in GUIDED, fly about 80 m from home, and switch to LOITER. **Write your prediction first.**

Event	How to trigger it	Parameter(s) that decide the response	Prediction	Observed mode, time to react, landing point
D1 Radio loss	Set `SIM_RC_FAIL` = 1 (no RC pulses). Set back to 0 afterwards.	`FS_THR_ENABLE` (1 = RTL in most firmware)		
D2 Low battery	Check `BATT_LOW_VOLT` (for example 10.5 V). Set `BATT_FS_LOW_ACT` = 2 (RTL). Then set `SIM_BATT_VOLTAGE` below `BATT_LOW_VOLT`, for example 10.2.	`BATT_LOW_VOLT`, `BATT_LOW_TIMER`, `BATT_FS_LOW_ACT`		
D3 GPS loss	Set `SIM_GPS_DISABLE` = 1, or `SIM_GPS1_ENABLE` = 0 in newer firmware (search `SIM_GPS`). Restore afterwards.	`FS_EKF_ACTION`, `FS_EKF_THRESH`		

> **Notes.** The battery failsafe waits for the voltage to stay low for `BATT_LOW_TIMER` seconds, so expect a delay. After GPS loss, Loiter and RTL cannot hold position; watch which mode the autopilot falls back to and read the **Messages** tab. Reset every `SIM_` parameter you changed before the next event, and disarm between events if the vehicle lands.

Checkpoint D. For each event, was the response what the parameters predicted? If not, which parameter explains the difference?

8. Part E: Download the log and analyze it (25 min)

1. Land and disarm. Logs are written while armed and closed on disarm.
2. In **Flight Data**, open the **DataFlash Logs** tab and click **Download DataFlash Log Via Mavlink**. Pick the most recent log (largest number) and save the `.bin` file.
3. Open **plot.ardupilot.org** and drag the `.bin` file onto the page. The log is processed in your browser.
4. Plot the signals below and answer the worksheet. Use the mode bands at the top of the plot to align events.

Message.field	What it shows	Look for
`MODE`	Every mode change and its reason	The mode the failsafe chose, and when
`CTUN.Alt`, `CTUN.DAlt`	Estimated and desired altitude	The RTL climb to 30 m; the descent in Stabilize
`CTUN.ThO`	Throttle output (0 to 1)	Saturation near 1 or 0
`BAT.Volt`, `BAT.Curr`	Battery voltage and current	Your D2 voltage step; current during climb
`GPS.NSats`, `GPS.HDop`	Satellites and dilution of precision	The D3 dropout
`ATT.Roll`, `ATT.DesRoll`	Actual and desired roll	How closely the controller tracks
`VIBE.VibeX/Y/Z`	Vibration levels	Should be low in simulation; on real aircraft above about 30 m/s² is a warning sign
ERR, EV	Failsafe and event codes	Radio, battery, fence and EKF entries

9. Log-analysis worksheet

Fill this in from your own log, not from memory.

#	Question	Your answer, with value and time stamp
1	Time from `SIM_RC_FAIL` = 1 to the mode change in `MODE`. Which mode?	
2	Peak of `CTUN.Alt` during the fence-triggered RTL. Does it match `RTL_ALT`?	
3	`BAT.Volt` just before and just after the D2 step. Time from step to failsafe. Compare with `BATT_LOW_TIMER`.	
4	Minimum `GPS.NSats` during D3, and the mode the vehicle ended in.	
5	Largest gap between `CTUN.Alt` and `CTUN.DAlt` in the whole flight. In which mode, and why?	
6	Was `CTUN.ThO` ever saturated? When?	
7	List every `ERR` entry with its subsystem and time.	
8	Sketch (or screenshot) one plot that best tells the story of your flight, and annotate three events on it.	

10. Viva questions

1. Why does Loiter need GPS while AltHold does not? What does each mode use to hold altitude?
2. Your RTL altitude is 15 m and the site has a 22 m tree between the work area and home. What happens on radio loss, and what would you change?
3. What is the difference between a geofence breach action and a failsafe action? Can both happen in one flight?
4. Why is the battery failsafe based on voltage held for several seconds rather than an instant reading?
5. After GPS loss, why can't the vehicle simply fly RTL? What can it still do safely?
6. Which log messages would you use to tell a vibration problem from a tuning problem?
7. Why are `RTL_ALT` and `FENCE_ALT_MAX` given in different units, and how would you protect against that mistake on a real aircraft?
8. In simulation, failsafes are free. Which of today's tests would you *never* do on a real aircraft over people, and how would you test it instead?
9. What does `CTUN.ThO` near 1.0 for several seconds tell you about the aircraft's design or payload?
10. How is SITL different from hardware-in-the-loop (HITL) and from a real flight? Name one thing SITL cannot show you.

11. What to submit

- The completed worksheet (sections 5 to 9), one per group.
- One annotated log plot.
- One paragraph: the failsafe priority you would set for a 2 kg survey drone, with reasons.

12. Instructor notes

- **Timing:** A 15, B 25, C 20, D 25, E 25, buffer 10 minutes. If time is short, do D1 and D2 only.
- **Common snags:** pre-arm messages before GPS fix (wait); parameters not saved (click Write Params, then Refresh); failsafe not firing because the matching `_ENABLE` or `_ACT` parameter is 0; students forgetting to reset `SIM_` parameters.
- **Hedge on names:** parameter names above are from recent ArduCopter 4.x releases. Check them against the version Mission Planner downloads before the class and correct this sheet if any differ.

- **Extension for fast groups:** plan a five-waypoint Auto mission in **PLAN**, write it, take off in GUIDED, switch to AUTO, and trigger D1 halfway. Compare RTL with FS_THR_ENABLE set to continue the mission in Auto (check the option list for your firmware).